INDIAN INSTITUTE OF TECHNOLOGY (ISM) DHANBAD

DEPARTMENT OF ENVIRONMENTAL SCIENCE AND ENGINEERING

End-Semester Examination (Monsoon Semester: 2025-2026)

Environmental Sciences (NESV101)

Date: 23.11.2025 **Duration:** 2 hour

Time: 1:30 PM to 3:30 PM **Total Marks:** 50

Note: **All questions are compulsory and Please answer the questions in order**

Unit 1

Question 1a: Draw an environmental impact assessment (EIA) process flowchart for a Category A project. [2.5 Marks]

Question 1b: Determine the amount of air required (by volume) per day for the combustion of municipal solid waste with molecular formula: $C_{800}H_{2180}O_{923}N_{13}S$. The waste feed rate is 400 tonnes per day (on a dry basis), and the excess air requirement is 150% per tonne of waste. Air contains 23% oxygen by weight, has a density of 1.29 kg/m^3. [5 Marks]

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1409369996

(1)

Question 1c: A city generates municipal solid waste at a rate of 0.8 kg per capita per day, with a population of 2.8 million. The city plans to provide land for the landfilling of waste generated over a 5-year period. The density of compacted waste in the landfill is 500 kg/m^3, and the landfill lift is 15 m. If 25% of the total landfill volume is occupied by soil cover, calculate the land area required for the landfill. [2.5 Marks]

?

7.267×10^5

Question 1d: What is a transfer station? Explain the economic feasibility of a transfer station using cost–distance curve. [2.5 Marks]

Unit 2

Question 2a: Differentiate between exponential and logistic population growth. Plot the graph and give the generic equation for both. [3 Marks]

Question 2b: A population of bacteria can be represented by the following logistic model:

$$\frac{dN}{dT} = 0.1N\left(1 - \frac{N}{300}\right)$$

X (i) What is the carrying capacity? (ii) If N (0) = 50, find the initial growth rate. [3 Marks]

(8)

X **Question 2c:** Highlight four differences between aerobic and anaerobic biological treatment of wastewater. [4 Marks]

Question 2d: A water treatment plant receives 12,000 m^3/day of raw water. A plain sedimentation tank requires a detention time of 4 hours. If the depth of the tank is 3 m, find the surface area required. [2.5 Marks]

$CH_4 + O_2 \longrightarrow CO_2 + H_2O$

$C + O_2 \rightarrow CO_2$

12 32

P.T.O

$\frac{800}{12} \times 32$

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Unit 3

Question 4a: Explain with chemical reactions, the carbon sequestration process taking place in the Ocean. **[5 Marks]**

Question 4b: Explain the simple global temperature model and estimate the effective temperature of the earth using the model. **[5 Marks]**

Question 4c: With a neat sketch, elucidate the global atmospheric general circulation. **[2.5 marks]**

Unit 4

Question 5a: *Four noise level readings were taken* at an industrial site at different times of the day as follows, calculate the Average Sound Pressure Level ($\bar{L}_p$) in dB (re = 20 μPa). **[3.5 Marks]**

S. No.	Time Interval	Sound Level (dB)
1	9:00–9:15 AM	48 dB
2	11:00–11:30 AM	60 dB
3	2:00–2:05 PM	70 dB
4	4:00–4:10 PM	64 dB

Question 5b: A thermal power plant of 500 MW capacity emits pollutants as follows, Calculate the required stack height (in m) for safe dispersion of pollutants as per CPCB guidelines/formula/method. **[3 Marks]**

10.3

- Particulate matter (PM) = 10 tonnes/day use h_1
- Sulphur dioxide (SO_2) = 600 kg/hr h_2

Question 5c: List three major Acts or Rules each under the following categories of Environmental Acts and Rules in India: **[3 Marks]**

(i) Pollution ~~Population~~ Control Acts
(ii) Conservation Acts
(iii) Institutional / Legal Acts

Question 5d: Explain the Chapman mechanism for the formation and natural cycling of ozone in the stratosphere. Write the main chemical reactions involved and state why the Chapman cycle alone cannot explain observed ozone losses. **[3 Marks]**

Formula
$h_1 = 74(Q_p)^{0.27}, \quad h_2 = 14(Q_s)^{1/3}$
$\overline{L_p} = 20\log_{10}\left(\frac{1}{N}\sum_{n=1}^{N} 10^{L_n/20}\right), \quad L_{eq} = 10\log_{10}\left(\sum_{i=1}^{n} 10^{L_i/10} \times t_i\right)$

$$\frac{10^{2.4}}{4} + \frac{10^{3}}{2} + \frac{10^{3.5}}{12} + \frac{10^{3.2}}{6} =$$

$C + O_2 \rightarrow CO_2$

$2H_2 + O_2 \rightarrow 2H_2O$

4 32